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Trunking Quiz

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Result of VLAN & Trunking Quiz:

Total Questions	Full Score	Passing Rate	Your Score	Correct Answer Percentage	Elapsed
10	100	80%	100	100%	00:02:23

Congratulations!

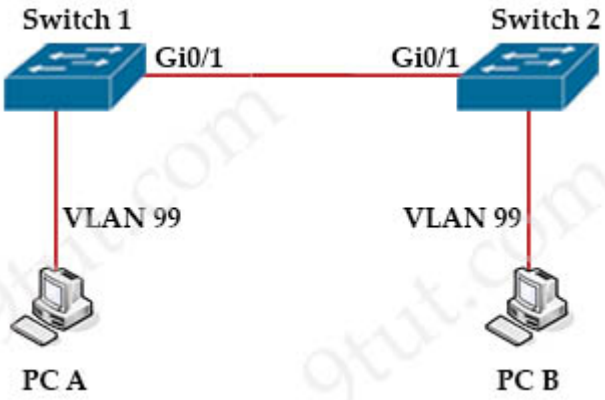
You passed this test!

If you want to retake this quiz, please press Ctrl + F5 on Windows or press CMD + R on Mac.

Your answers are shown below:

Question 1

Refer to the exhibit. After the switch configuration the ping test fails between PC A and PC B. Based on the output for switch 1, which error must be corrected?



Switch 1	**Switch 2**
Name: Gi0/1	Name: Gi0/1
Switchport: Enabled	Switchport: Enabled
Administrative Mode: trunk	Administrative Mode: trunk
Operational Mode: trunk	Operational Mode: trunk
Administrative Trunking Encapsulation: dot1q	Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: dot1q	Operational Trunking Encapsulation: dot1q
Negotiation of Trunking: Off	Negotiation of Trunking: Off
Access Mode VLAN: 1 (default)	Access Mode VLAN: 1 (default)
Trunking Native Mode VLAN: 1 (default)	Trunking Native Mode VLAN: 99 (default)

Administrative Native VLAN tagging: enabled Voice VLAN: none <output omitted> Trunking VLANs Enabled: 50-100 Pruning VLANs Enabled: 2-1001 Capture Mode Disabled Capture VLANs Allowed: ALL	Administrative Native VLAN tagging: enabled Voice VLAN: none <output omitted> Trunking VLANs Enabled: 50-100 Pruning VLANs Enabled: 2-1001 Capture Mode Disabled Capture VLANs Allowed: ALL

- **A. There is a native VLAN mismatch**
- B. Access mode is configured on the switch ports.
- C. The PCs are in the incorrect VLAN
- D. All VLANs are not enabled on the trunk

Explanation

From the output we see the native VLAN of Switch1 on Gi0/1 interface is VLAN 1 while that of Switch2 is VLAN 99 so there would be a native VLAN mismatch.

Question 2

Two switches are connected and using Cisco Dynamic Trunking Protocol. SW1 is set to Dynamic Desirable. What is the result of this configuration?

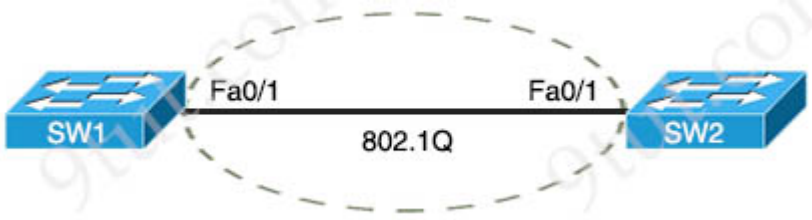
- A. The link is in a downstate.
- B. The link is in an error disables stale
- C. The link is becomes an access port
- **D. The link becomes a trunkport**

Explanation

Maybe this question is missing the “SW2 is set to Dynamic Auto” part so we assume this part to find out the best answer. Dynamic Desirable + Dynamic Desirable/Dynamic Auto/Trunk will form a trunk link.

Question 3

Refer to exhibit. Which action do the switches take on the trunk link?



interface FastEthernet0/1 switchport trunk encapsulation dot1q switchport trunk native vlan 999 switchport mode trunk	interface FastEthernet0/1 switchport trunk encapsulation dot1q switchport trunk native vlan 99 switchport mode trunk

- A. The trunk does not form and the ports go into an err-disabled status

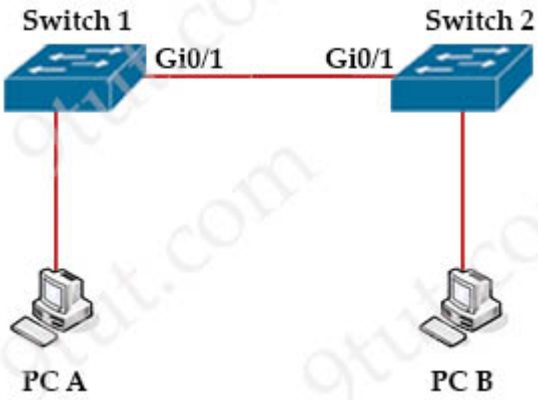
- **B. The trunk forms but the mismatched native VLANs are merged into a single broadcast domain**
- C. The trunk does not form, but VLAN 99 and VLAN 999 are allowed to traverse the link
- D. The trunk forms but VLAN 99 and VLAN 999 are in a shutdown state

Explanation

The trunk still forms with mismatched native VLANs and the traffic can actually flow between mismatched switches. But it is absolutely necessary that the native VLANs on both ends of a trunk link match; otherwise a native VLAN mismatch occurs, causing the two VLANs to effectively merge. For example with the above configuration, SW1 would send untagged frames for VLAN 999. SW2 receives them but would think they are for VLAN 99 so we can say these two VLANs are merged.

Question 4

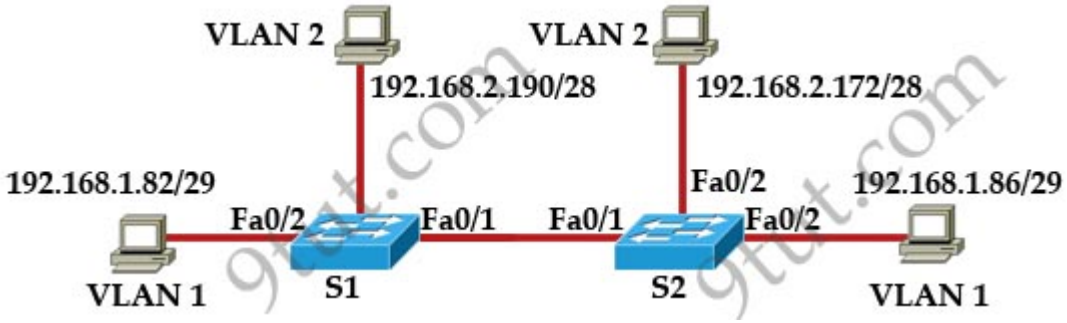
Refer to the exhibit. The network administrator wants VLAN 67 traffic to be untagged between Switch 1 and Switch 2 while all other VLANs are to remain tagged. Which command accomplishes this task?



- A. switchport access vlan 67
- B. switchport trunk allowed vlan 67
- C. switchport private-vlan association host 67
- **D. switchport trunk native vlan 67**

Question 5

Refer to the exhibit.



```
S1#show interface trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802. 1q        Trunking      1

Port        Vlans allowed a trunk
Fa0/1       1.1005

Port        Vlans allowed and active in management domain
Fa0/1       12
```

```
S2#show interface trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802. 1q        Trunking      2

Port        Vlans allowed a trunk
Fa0/1       1.1005

Port        Vlans allowed and active in management domain
Fa0/1       12
```

A frame from VLAN1 of switch S1 is sent to switch S2 where the frame received on VLAN2. What causes this behavior?

- A. trunk mode mismatches
- B. VLANs that do not correspond to a unique IP subnet
- **C. native vlan mismatches**
- D. allowing only vlan 2 on the destination

Explanation

For 802.1q encapsulation, the native VLAN must matched at both side; otherwise the link will not work. In this case the native VLAN of S1 is 1 while the native VLAN of S2 is 2. Therefore when a frame from VLAN 1 is sent from S1 to S2, it will be untagged before leaving S1. When S2 receives this untagged frame, it believes this frame was sent from the native VLAN. But the native VLAN on S2 is VLAN 2, not VLAN 1. Therefore only hosts on VLAN2 of S2 will receive this frame.

Question 6

Refer to the exhibit. Which action is expected from SW1 when the untagged frame is received on the GigabitEthernet0/1 interface?

```
SW1#show run int gig 0/1
interface GigabitEthernet0/1
 switchport access vlan 11
 switchport trunk allowed vlan 1-10
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 5
 switchport mode trunk
 speed 1000
 duplex full
```

- **A. The frame is processed in VLAN 5**
- B. The frame is processed in VLAN 11
- C. The frame is processed in VLAN 1

- D. The frame is dropped

Question 7

Refer to the exhibit.



SW1

```
interface FastEthernet 0/1
 channel-group 1 mode auto
 switchport trunk encapsulation dot1q
 switchport mode trunk

interface FastEthernet 0/2
 channel-group 1 mode auto
 switchport trunk encapsulation dot1q
 switchport mode trunk
```

A network administrator is configuring an EtherChannel between SW1 and SW2. The SW1 configuration is shown. What is the correct configuration for SW2?

- A. interface FastEthernet 0/1
  channel-group 1 mode active
  switchport trunk encapsulation dot1q
  switchport mode trunk
  !
  interface FastEthernet 0/2
  channel-group 1 mode active
  switchport trunk encapsulation dot1q
  switchport mode trunk

- B. interface FastEthernet 0/1
  channel-group 2 mode auto
  switchport trunk encapsulation dot1q
  switchport mode trunk
  !
  interface FastEthernet 0/2
  channel-group 2 mode auto
  switchport trunk encapsulation dot1q
  switchport mode trunk

- **C. interface FastEthernet 0/1**
  channel-group 1 mode desirable
  switchport trunk encapsulation dot1q
  switchport mode trunk
  !
  interface FastEthernet 0/2
  channel-group 1 mode desirable
  switchport trunk encapsulation dot1q
  switchport mode trunk

- D. interface FastEthernet 0/1
  channel-group 1 mode passive
  switchport trunk encapsulation dot1q
  switchport mode trunk
  !
  interface FastEthernet 0/2
  channel-group 1 mode passive
  switchport trunk encapsulation dot1q
  switchport mode trunk

Explanation

From the configuration of SW1, we see it is using Port Aggregation Protocol (PAgP) with “auto” mode so the other end (SW2) must also runs PAgP with “desirable” mode to actively sending request to form an Etherchannel.

Question 8

What occurs to frames during the process of frame flooding?

- A. Frames are sent to all ports, including those that are assigned to other VLANs
- B. Frames are sent to every port on the switch that has a matching entry in MAC address table
- **C. Frames are sent to every port on the switch in the same VLAN except from the originating port**
- D. Frames are sent to every port on the switch in the same VLAN

Question 9

Refer to the exhibit. An administrator is tasked with configuring a voice VLAN. What is the expected outcome when a Cisco phone is connected to the GigabitEthernet 3/1/4 port on a switch?

```
interface GigabitEthernet3/1/4
switchport voice vlan 50
!
```

- A. The phone and a workstation that is connected to the phone do not have VLAN connectivity.
- **B. The phone sends and receives data in VLAN 50, but a workstation connected to the phone sends and receives data in VLAN 1.**
- C. The phone sends and receives data in VLAN 50, but a workstation connected to the phone has no VLAN connected.
- D. The phone and a workstation that is connected to the phone send and receive data in VLAN 50.

Question 10

Refer to the exhibit. What commands are needed to add a subinterface to Ethernet0/0 on R1 to allow for VLAN 20, with IP address 10.20.20.1/24?



R1 interface Ethernet0/0 no ip address	SW1: interface Ethernet0/0 switchport trunk encapsulation dot1q switchport mode trunk ! interface Ethernet0/1 switchport trunk allowed vlan 10 switchport trunk encapsulation dot1q switchport mode trunk	SW2: interface Ethernet0/1 switchport trunk encapsulation dot1q switchport mode trunk ! interface Ethernet0/2 switchport access vlan 20 switchport mode access

- A. R1 (config)#interface ethernet0/0
  R1 (config)#encapsulation dot1q 20
  R1(config)#ip address 10.20.20.1 255.255.255.0
- **B. R1 (config)#interface ethernet0/0.20**
  R1 (config)#encapsulation dot1q 20
  R1(config)#ip address 10.20.20.1 255.255.255.0
- C. R1 (config)#interface ethernet0/0.20
  R1(config)#ip address 10.20.20.1 255.255.255.0
- D. R1 (config)#interface ethernet0/0
  R1(config)#ip address 10.20.20.1 255.255.255.0

Comments (19) Comments

1. ILIYA
   July 21st, 2020

   Question 3….Explanation makes no sense…what? SW1 would send untagged frames for VLAN999? I thought SW1 has the VLAN999 settings and SW2 has VLAN99 settings, or is there a typo in the explanation ?

   Appreciate the clarification.

2. Darkonzy
   August 6th, 2020

   Iliya, native vlans are used to send untagged frames. Therefore, SW1 would send untagged frames on its native VLAN999, and since the frames are untagged, SW2 would receive them on VLAN99, since VLAN99 is its native VLAN.

3. Andrew
   February 6th, 2021

   Hi, can you please explain question 10 why does .20 need to be added after interface ethernet0/0 ?

4. Mohsen
   February 7th, 2021

   Andrew, it is because we have at least two vlans here, Vlan 10 and Vlan 20 and they have different network addresses and need different gateway addresses, but only on einterface on the router so we need to have subinterfaces configured on the router. ethernet 0/0.20 for vlan 20 and eternet 0/0.10 for vlan 10 each with different ip addresses

5. Neev

February 15th, 2021

9tut, please explain

Refer to the exhibit. An administrator is tasked with configuring a voice VLAN. What is the expected outcome when a Cisco phone is connected to the GigabitEthernet 3/1/4 port on a switch?

interface GigabitEthernet3/1/4
switchport voice vlan 50
!
A. The phone and a workstation that is connected to the phone do not have VLAN connectivity.
B. The phone sends and receives data in VLAN 50, but a workstation connected to the phone sends and receives data in VLAN 1.correct
C. The phone sends and receives data in VLAN 50, but a workstation connected to the phone has no VLAN connected.
D. The phone and a workstation that is connected to the phone send and receive data in VLAN 50.

6. jamaal
February 28th, 2021

best of best questions thanks 9tut

7. anon
May 26th, 2021

@Neev This configuration is used when you want to daisy chain a phone to a PC. IP phones have a built in 3 way switch Usually ‘LAN’ and ‘PC’ – This allows the phone to have priority over VoIP packets whilst the data packets can be forwarded to the PC from the phone. So the PC will use VLAN 1 and the and the IP phone will use the VLAN configured on the switchport interface

8. Anonymous
May 26th, 2021

@Andrew
This is used as an alternative to Inter-vlan routing (it’s not the best method, traditionally you’d want to use a multi-layer switch). A Router can mediate traffic between multiple vlans by configuring the router interface as a sub-interface. This interface must be configured as a trunk (for obvious reasons… sending tagged frames). to signify the sub interface you use (example g0/0/0.20 (the .xx value doesnt actually need to represent the vlan – it’s just convenient) and then in config-if encapsulation do1q Nice and simple

9. anishakya
January 28th, 2022

@Andrew @9tut
on Q10, does the vlan20 can reach the router sub interface? SW1 E0/1 connecting to SW2 has allowed vlan 10 defined.

10. Jacob B
February 22nd, 2022

Done

11. anon123
July 18th, 2022

Hello, question 4 does not make sense. It wants you to untag the VLAN 67 between the two switches but then the question only gives you the options to trunk the VLAN or make the port access it.

12. Yunshan
July 25th, 2022

@anon123
The function of native vlan is to create an untagged frame link so the answer D. is correct

13. Q2
September 19th, 2022

Q2 – the question should be: Two switches are connected and using Cisco Dynamic Trunking Protocol. SW1 is set to Dynamic Auto and SW2 is set to Dynamic Desirable. What is the result of this configuration?

14. exam
September 21st, 2022

Has anyone taking the most recent exam? Are there any lab sim questions?

15. Explanation To A Question
December 28th, 2022

Refer to the exhibit. An administrator is tasked with configuring a voice VLAN. What is the expected outcome when a Cisco phone is connected to the GigabitEthernet 3/1/4 port on a switch?

interface GigabitEthernet3/1/4
switchport voice vlan 50

The computer will be in a data VLAN, the IP phone will be in the voice VLAN.
…
Behind the scenes, we have a trunk between our switch and IP phone. The port on the IP phone that connects to the computer is an access port. The IP phone will forward all traffic from the computer to the switch untagged, traffic from the IP phone itself will be tagged. The only two VLANs that are allowed though, are the access and voice VLAN.

The phone sends and receives data in VLAN 50, but a workstation connected to the phone sends and receives data in VLAN 1.

Correct answer.

16. sure°°
October 16th, 2023

no comments 2023?

17. not sure°°
September 23rd, 2024

no comments 2024?

18. not not sure
September 19th, 2025

no comments 2025?

19. Rv
August 17th, 2026

is this still valid?

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- === New CCNA v1.1 Lab Sims (very important) ===
- Voice VLAN Configuration Sim 2
- Voice VLAN & LLDP Sim
- Allowed, Native VLAN & LACP Sim
- Static Routing Configuration Sim 4
- IPv4 and IPv6 Connectivity Sim
- VLAN & CDP Sim 2
- IPv4 and IPv6 Assignment Sim
- VLAN CDP & LLDP Sim
- OSPF Configuration Sim 2
- OSPF Configuration Sim 3
- Trunking & LACP Sim
- VLAN Configuration Sim
- Allowed, Native VLAN & LACP Sim 2
- 802.1Q Trunking & LACP
- IPv4 and IPv6 Assignment Sim 2
- 802.1Q Trunking, Native VLAN & LACP
- Static Routing Configuration Sim 5
- Static Routing Configuration Sim 6
- PAT, NTP & DHCP Relay Sim
- OSPF DR Sim
- NACL DHCP Snooping 3
- NAT DHCP SSH Sim
- ==== Lab Sims for CCNA (although they are rather old but please learn all of them too) ====
- IP Services Sim Version 2
- OSPF Configuration Sim
- LACP Configuration Sim

- Voice VLAN Configuration Sim
- VLAN and Trunking Configuration Sim
- Named Access-list & Port Security Sim
- Named Access-list & DHCP Snooping Sim
- VLAN and CDP Sim
- IPv4 and IPv6 Deployment Sim
- Static Routing Configuration Sim 3
- VLAN & LLDP Sim
- IPv6 Static Routing Sim
- Named Access-list & DHCP Snooping Sim 2
- ISL Trunking and LACP Sim
- === Below Sims are old and retired so you do not need to learn them ===
- Static Routing Configuration Sim 2
- IP Services Sim
- Static Routing Configuration Sim

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